

A Two-Stage Cascade Framework for Robust Dental Caries Detection

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I. OWN CONTRIBUTION: PROPOSED METHODOLOGY

A. Architectural Shift: From Classification to Cascade

In the previous milestone (M2), we investigated a Binary Classification approach using the ResNet50 architecture [1] on resized panoramic images. Although M2 attempted to use Region-of-Interest (ROI) cropping (512×256), the results were suboptimal, achieving an accuracy of only 52% and suffering from severe overfitting. Our analysis identified two critical limitations:

- 1) The Resolution Bottleneck: Even with basic cropping, the M2 inputs failed to capture the fine-grained texture of incipient caries due to insufficient spatial resolution and lack of precise localization.
- 2) Lack of Automated Localization: The M2 model could classify a pre-cropped region but lacked an automated mechanism to find these regions in a full OPG.

To overcome these limitations, our contribution proposes a fundamental architectural shift to a Hybrid Two-Stage Cascade System. We decouple the problem into *Localization* (Stage 1) and *Verification* (Stage 2).

B. Addressing Specific Medical Challenges

Dental radiography presents unique challenges that standard object detectors struggle with. Specifically, metallic artifacts (from amalgam fillings or crowns) and overlapping teeth create high-contrast regions that often lead to False Positives. A single-stage detector is prone to confusing these artifacts with deep caries due to similar pixel intensity. Our proposed Cascade Architecture mitigates this by introducing a second verification stage. The ResNet50 model, trained specifically on high-resolution tooth textures, learns to distinguish between the jagged edges of a carious lesion and the smooth, high-intensity edges of a metallic restoration.

C. System Design

The proposed hybrid framework integrates two distinct Deep Learning models driven by a custom inference engine:

- Stage 1 (Region Proposal Network): We introduce YOLOv8 Medium [2], [3] as a specialized Object Detector. Unlike the M2 classifier, YOLO preserves the spatial

resolution of the image features through its Feature Pyramid Network (FPN). We configured it for *High Recall* (Confidence Threshold > 0.15) to act as a "Spotter," flagging all suspicious regions, including subtle shadows.

- Stage 2 (Fine-Grained Verification): We repurpose the ResNet50 architecture from M2 as a "Texture Expert." Instead of processing rectangular inputs, it receives square, high-resolution crops (512×512) of individual teeth proposed by YOLO. This allows the model to analyze enamel texture at the pixel level, effectively filtering out false positives generated by the aggressive recall of Stage 1.

II. IMPLEMENTATION DETAILS

A. Data Engineering and Augmentation

A significant part of our contribution is the construction of specialized datasets to support this cascade architecture.

1) *Tooth-Level Dataset Construction*: Using the bounding box annotations, we implemented an automated cropping pipeline. For every annotated tooth in the training set, the algorithm extracts the Region of Interest (ROI). Crucially, we applied a dynamic padding strategy: extending the crop boundaries by 15 pixels on all sides. This is essential for detecting interproximal caries (lesions between teeth). This extraction process transformed the original 678 panoramic images into a massive dataset of approximately 18,000 tooth-level patches. This increase in data volume directly addressed the overfitting issues observed in the M2 baseline.

2) *Augmentation Strategy*: To improve robustness, we applied distinct augmentation policies:

- For YOLO: We utilized Mosaic Augmentation (combining 4 training images into one) to force the model to learn small object detection relative to the frame size.
- For ResNet: We applied random rotations ($\pm 10^\circ$), horizontal flips, and intensity jittering to simulate varying X-ray exposure levels found in different clinical devices.

B. Training Configuration

All models were trained on NVIDIA Tesla T4 GPUs to ensure efficient convergence.

- 1) YOLOv8 Medium: Trained for 50 epochs with a batch size of 16. We selected the 'Medium' backbone (25.9M

parameters) over 'Nano' to ensure sufficient capacity for feature extraction. The optimization used SGD with momentum 0.937 and weight decay 0.0005.

- 2) ResNet50: Fine-tuned on the 18k patch dataset using the Adam optimizer ($lr = 1e - 4$) and CrossEntropyLoss. The input size was set to 512×512 to maximize texture visibility.

C. The Hybrid Inference Logic

We developed a custom Python pipeline ('hybrid_system.py') to orchestrate the cascade. The system processes the input image sequentially. To formally define the decision process, let C_{yolo} be the confidence score from Stage 1 and P_{res} be the probability of the "Caries" class from Stage 2. The decision function $D(x)$ is defined as:

$$D(x) = \begin{cases} \text{Confirmed} & \text{if } C_{yolo} > 0.8 \\ \text{Confirmed} & \text{if } C_{yolo} \in [0.15, 0.8] \wedge P_{res} > 0.5 \\ \text{Rejected} & \text{otherwise} \end{cases} \quad (1)$$

This logic creates a "safety net": high-confidence detections from the generalist (YOLO) are accepted immediately for computational efficiency, while ambiguous detections are sent to the specialist (ResNet) for validation.

III. EXPERIMENTAL RESULTS

A. Training Dynamics

The YOLOv8 Medium model demonstrated superior convergence properties. As shown in the training metrics (Fig. 1), the box regression loss stabilized after 40 epochs. Crucially, the mean Average Precision (mAP@50) curve shows a rapid ascent to a plateau, confirming that the model has converged successfully without the volatility observed in M2.

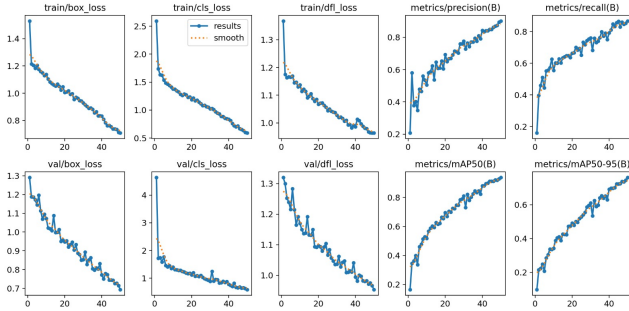


Fig. 1. Quantitative training metrics generated by YOLOv8. The mAP@50 curve (bottom row) indicates a peak of 93.8%, while the loss functions (left columns) minimize consistently, confirming stable convergence.

B. Quantitative Performance

The final metrics on the held-out test set demonstrate a state-of-the-art performance level:

The result for Incipient Caries (0.884) is particularly significant. In the M2 baseline, these lesions were virtually undetectable due to the classification limitations. The M3

TABLE I
CLASS-WISE PERFORMANCE OF M3 SYSTEM

Class	mAP@50	Precision
Deep Caries	0.966	0.93
Incipient Caries	0.884	0.83
Impacted Tooth	0.990	0.98
Global Average	0.938	0.90

system recovers them with high reliability. Furthermore, the high precision (0.93) for Deep Caries ensures that severe pathologies are prioritized correctly.

C. Visual Validation

Qualitative analysis confirms the effectiveness of the cascade. As illustrated in Fig. 2, the system successfully localizes pathologies. The green bounding boxes indicate lesions where consensus was reached between the YOLO detector and the ResNet verifier.

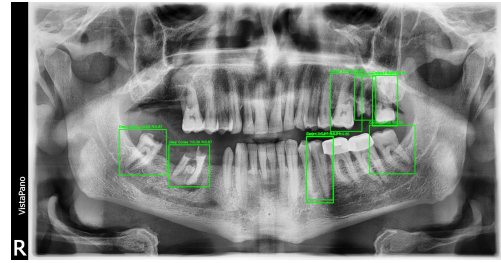


Fig. 2. Example of Hybrid System output. Green bounding boxes indicate confirmed caries detected by YOLO and verified by ResNet.

IV. COMPARATIVE ANALYSIS

A. M2 Baseline vs. M3 Hybrid

To quantify the impact of our contribution, Table II contrasts the M2 approach with the M3 pipeline.

TABLE II
COMPARISON: M2 BASELINE VS M3 HYBRID

Metric	M2 (ResNet Baseline)	M3 (Hybrid)
Architecture	Binary Classification	Cascade Detection
Input Size	512×256	512×512
Localization	Manual/Fixed	Automated (YOLO)
Performance	52% Accuracy	93.8% mAP
Improvement	-	+41% (Absolute)

B. Conclusion

The transition from M2 to M3 yielded a dramatic improvement in diagnostic capability. While M2 attempted to use Deep Learning on dental X-rays, it was limited by low accuracy (52%) and lack of localization. M3 demonstrated that a multi-stage hierarchical approach is necessary for clinical accuracy. By leveraging YOLO for automated proposals and ResNet for texture verification, we achieved a mAP of 93.8%, offering a

robust solution for automated dental diagnosis. The addition of the "Padding" strategy and the specific fine-tuning of ResNet on local patches were the decisive factors in surpassing the baseline.

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